BTEC Level 3 Applied Science · Unit 1 Chemistry · Answer sheet

Atomic Structure — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

A

1. Proton +1, mass 1. Neutron 0, mass 1. Electron −1, mass ≈ 1/1836.
2. Z = number of protons. A = protons + neutrons.
3. Same proton number / element; different neutron number / mass number.

B

4. 19 p, 20 n, 19 e.
5. Number of protons equals number of electrons, so charges cancel.

C

6. Mg: 12p 12n 12e. Mg^{2+}: 12p 12n 10e. F: 9p 10n 9e. F^{-}: 9p 10n 10e.
7. $A_r = (60\times69 + 40\times71)/100 = 69.8$
8. Chlorine is a mixture of ^{35}Cl and ^{37}Cl; A_r is a weighted mean.

D / E

9. Isotopes must have the same proton number. C has 6 protons; N has 7.
10. $(75\times63 + 25\times65)/100 = 63.5$
11. The atom gains two electrons. Proton number unchanged. Electrons exceed protons by 2, so charge is 2−. [3]
12. $A_r = (90.5\times20 + 0.27\times21 + 9.25\times22)/100 = 20.2$ (3 s.f.). Ignoring ^{21}Ne changes only the third significant figure.
13. A = 23 top left, Z = 11 bottom left, 11 p, 12 n. Na^{+} has 10 electrons.